

Hamiltonian Boundary Value Method HBVM(4,2)

Rank-two polynomial path with four Gauss nodes

1 Polynomial-path formulation

Let $(c_i, b_i)_{i=1}^4$ be four-point Gauss–Legendre quadrature on $[0, 1]$. For the first two orthonormal shifted Legendre polynomials,

$$P_0(c) = 1, \quad P_1(c) = \sqrt{3}(2c - 1),$$

define

$$I_{i0} = c_i, \quad I_{i1} = \sqrt{3}(c_i^2 - c_i).$$

HBVM(4,2) solves for two vector coefficients γ_0, γ_1 :

$$Y_i = y_n + h \sum_{j=0}^1 I_{ij} \gamma_j, \tag{1}$$

$$R_j(\gamma) = \gamma_j - \sum_{i=1}^4 b_i P_j(c_i) f(t_n + c_i h, Y_i) = 0. \tag{2}$$

The accepted state is

$$y_{n+1} = y_n + h \sum_{i=1}^4 b_i f(t_n + c_i h, Y_i).$$

Equivalently, the Runge–Kutta matrix is $A = \mathcal{I}_2 \mathcal{P}_2^T \Omega_q$ and has rank two. The nonlinear unknown therefore has twice the state dimension, although four stages evaluate the line integral.

2 Order and energy property

With polynomial degree $s = 2$, the method has classical order $2s = 4$. Four-point Gauss quadrature has degree seven. For an autonomous polynomial Hamiltonian of degree ν , exact polynomial energy conservation follows when $k \geq \nu s/2$; hence HBVM(4,2) covers degrees through four. For a general smooth Hamiltonian the quadrature error makes energy conservation approximate, with an error that decreases rapidly with the step and quadrature accuracy.

HBVM(k, s) is generally not symplectic when $k > s$. Thus HBVM(4,2) must not be labeled symplectic by construction; map-Jacobian diagnostics measure its canonical defect independently of its energy behavior.

3 Nonlinear configurations

Newton acts on the reduced residual (R_0, R_1) . The public `jacobian_method` values are `analytic`, `finite_difference`, and `auto`. Analytic mode assembles exact guiding-centre particle blocks at the four stages; finite-difference mode differentiates the stage fields; auto selects the available capability. A backtracking correction is used only when the full Newton update increases the residual. The stopping scale is

$$\varepsilon_{\text{abs}} + \varepsilon_{\text{rel}} \max(1, \|y_n\|_\infty).$$

With `track_energy=True`, the implementation stores the physical Hamiltonian and its drift at saved times. For explicitly time-dependent systems this is a diagnostic, not an invariant claim.

4 Implementation correspondence

Coefficients, residuals, Newton assembly, and integration are in `src/methods/hbvm/order4.py`. The simulation document describes required Hamiltonian capabilities, work counters, fixed-grid sampling, and evaluation studies; shared physical dynamics are documented outside the model tree.

5 Four state formulations and passive energy tracking

For one planar particle the supported internal representations are

physical	$z = (x, y)$	$\mathbb{R}^2$,
physical with energy	(z, t, κ)	$\mathbb{R}^4$,
duplicated	$(u, v), \quad u, v \in \mathbb{R}^2$	$\mathbb{R}^4$,
duplicated with energy	(u, v, t, κ)	$\mathbb{R}^6$.

Classical methods use the physical rows; ABBA and BM4 use the duplicated rows. `track_energy` selects the energy variant. For N planar particles the respective dimensions are $2N$, $4N$, $4N$, and $6N$: one time entry and one momentum per particle. All time entries equal the integration time. Classical full-cyclotron runs instead use $4N$ or $6N$ entries. Accepted spatial copies coincide; only one physical copy is exported. Every coordinate block contains N values in component-major order. The formulation validates particle times and aligns output round-off; the integrator does not interpret coordinate indices.

The normalized passive balance is

$$\dot{\kappa} = -\partial_t H(t, z), \quad \kappa(t_0) = 0, \quad \Delta K(t) = H(t, z(t)) + \kappa(t) - H(t_0, z_0).$$

Splitting accumulators from both copies are divided by two. The physical Hamiltonian may vary with time; ΔK measures its computed balance. Tracking leaves physical trajectories, nonlinear stopping rules and adaptive acceptance unchanged. Dynamics must supply the momentum derivative, including zero for an autonomous Hamiltonian.

Hairer’s constraint is solely $u - v = 0$: the reduced multiplier has $2N$ entries and ABBA’s simultaneous spatial solve has $6N$ unknowns. Neither t nor κ is projected. The former eight-component projection is retired; `state_extension='fully_extended'` raises migration guidance.

Dynamics, formulation, method and integration remain separate. The shared state classes own layouts, diagonal embedding and history extraction in

`src/formulations/state.py`

Each integration initializes a fresh method instance. Observers receive physical states; energy histories are in `Solution.diagnostics`. The `extended_time` history has shape (N, n_{samples}) . Fixed-step shadow samples leave main states and counters unchanged. Adaptive energy quadrature uses accepted dense output; its backend state and Jacobian remain physical.

The four converged stage states and Gauss weights integrate the passive momentum. Its value never enters the Legendre coefficient solve.

6 Optional JAX execution

An optional JAX driver executes the same numerical map on one selected CPU or GPU, with particle operations batched inside a compiled sequential time loop. It reuses the canonical coefficients and prepared potential splines. Physical-state tolerance scaling, signed stage times, passive energy weights and independent shortened output maps retain their definitions above. Applicable Newton

systems use independent particle blocks; an explicitly selected Broyden iteration retains its global secant matrix. Nonlinear iteration limits and failure checks remain active on the device.

Both execution routes use double precision and return the same physical solution layout. Floating-point reassociation can produce round-off differences without changing the designed order or geometric properties. First-call compilation is separate from warmed integration timings, which include transfers and output construction. The sibling runtime document and shared execution contract specify supported callbacks and exact device scope.